



# **Design and Analysis of Algorithms**

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# DESIGN AND ANALYSIS OF ALGORITHMS

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## Basic Efficiency Classes

## Problems based on Asymptotic notations

Slides courtesy of **Anany Levitin**

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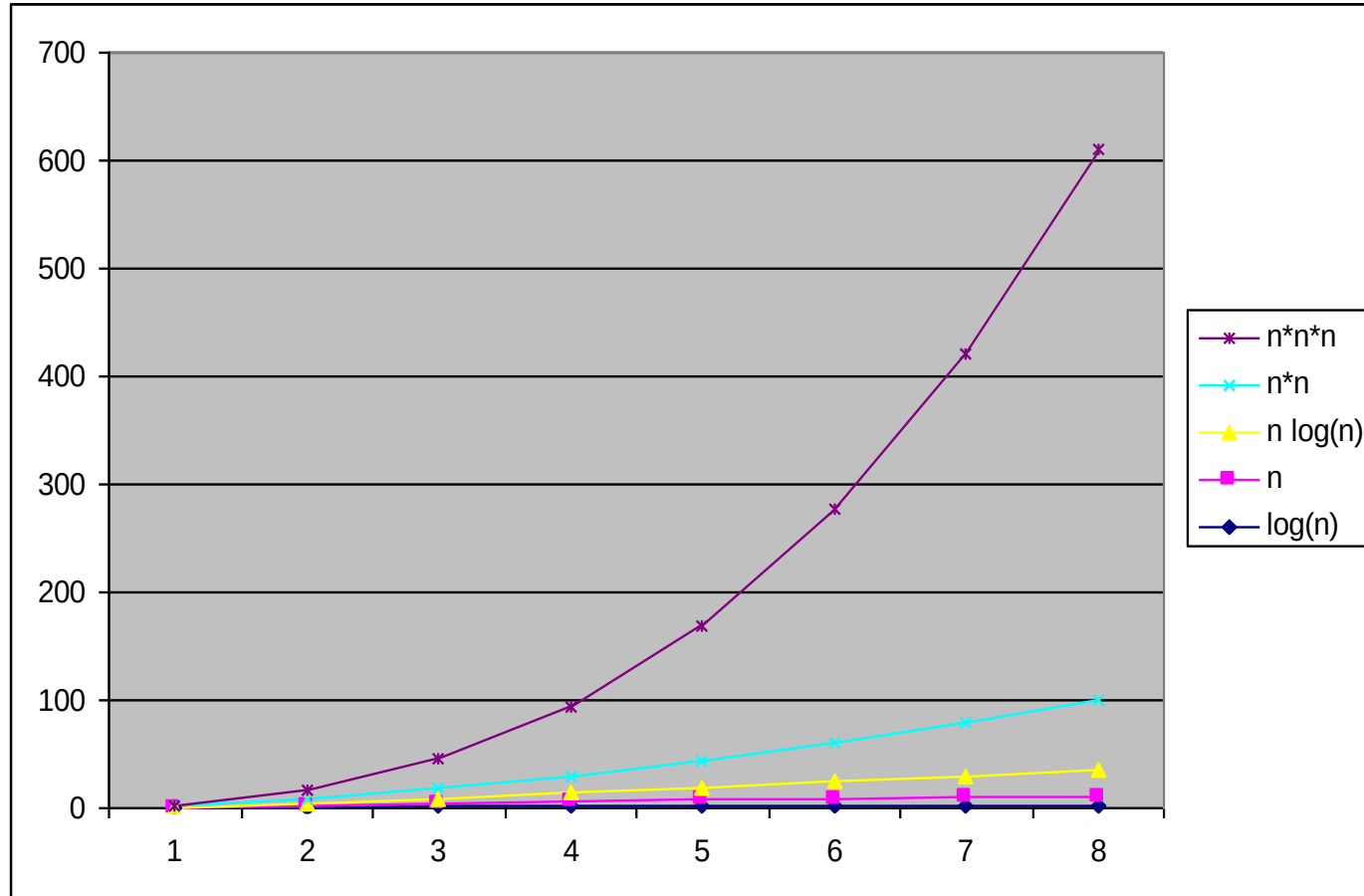
# Design and Analysis of Algorithms

## Basic Efficiency Classes

| Class      | Name          | Example                          |
|------------|---------------|----------------------------------|
| 1          | constant      | Best case for sequential search  |
| $\log n$   | logarithmic   | Binary Search                    |
| $n$        | linear        | Worst case for sequential search |
| $n \log n$ | $n$ -log- $n$ | Mergesort                        |
| $n^2$      | quadratic     | Bubble Sort                      |
| $n^3$      | cubic         | Matrix Multiplication            |
| $2^n$      | exponential   | Subset generation                |
| $n!$       | factorial     | TSP using exhaustive search      |

# Design and Analysis of Algorithms

## Basic Efficiency Classes



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## Basic Efficiency Classes

| $n$    | $\log_2 n$ | $n$    | $n \log_2 n$     | $n^2$     | $n^3$     | $2^n$               | $n!$                 |
|--------|------------|--------|------------------|-----------|-----------|---------------------|----------------------|
| 10     | 3.3        | $10^1$ | $3.3 \cdot 10^1$ | $10^2$    | $10^3$    | $10^3$              | $3.6 \cdot 10^6$     |
| $10^2$ | 6.6        | $10^2$ | $6.6 \cdot 10^2$ | $10^4$    | $10^6$    | $1.3 \cdot 10^{30}$ | $9.3 \cdot 10^{157}$ |
| $10^3$ | 10         | $10^3$ | $1.0 \cdot 10^4$ | $10^6$    | $10^9$    |                     |                      |
| $10^4$ | 13         | $10^4$ | $1.3 \cdot 10^5$ | $10^8$    | $10^{12}$ |                     |                      |
| $10^5$ | 17         | $10^5$ | $1.7 \cdot 10^6$ | $10^{10}$ | $10^{15}$ |                     |                      |
| $10^6$ | 20         | $10^6$ | $2.0 \cdot 10^7$ | $10^{12}$ | $10^{18}$ |                     |                      |

**Table 2.1** Values (some approximate) of several functions important for analysis of algorithms

# Design and Analysis of Algorithms

## Asymptotic notations

O

$$f(n) = 3n+2 \quad g(n) = n$$
$$3n+2 \in O(n)$$
$$f(n) \leq c g(n) \text{ for some } +ve \ c \ \forall n \geq n_0$$
$$\Rightarrow f(n) \in O(g(n))$$
$$3n+2 \leq cn$$

Let  $c = 4$

$$3n+2 \leq 4n$$

$n=1$   $n=2$   $n=3$

$5 \not\leq 4$   $8 \leq 8$   $11 \leq 12$

$3n+2 \leq 4n \quad \forall n \geq 2$

$$\Rightarrow 3n+2 \in O(n)$$

$\Omega$

$$3n+2 \in \Omega(n)$$
$$f(n) \geq c g(n) \text{ for some } +ve \ c \ \forall n \geq n_0$$
$$f(n) \in \Omega(n)$$
$$3n+2 \geq cn$$

Let  $c = 1$

$$3n+2 \geq n$$

$n=1$   $n=2$   $n=3$

$5 \geq 1$   $8 \geq 2$   $11 \geq 3$

$$3n+2 \geq n \quad \forall n \geq 1$$
$$3n+2 \in \Omega(n)$$

$\Theta$

$$3n+2 \in \Theta(n)$$
$$c_1 g(n) \leq f(n) \leq c_2 g(n) \quad \forall n \geq n_0$$

$c_1 = 1 \quad c_2 = 4$   
 $n_{01} = 1 \quad n_{02} = 2$

$n_0 = \max(n_{01}, n_{02}) = 2$

$c_1 = 1 \quad c_2 = 4 \quad n_0 = 2$

$$3n+2 \in \Theta(n)$$
$$f(n) \in O(g(n))$$
$$f(n) \in \Omega(g(n))$$
$$\Rightarrow f(n) \in \Theta(g(n))$$



# THANK YOU

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